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Home-built magnetic resonance imaging system (0.3 T) with a complete digital spectrometer

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A home-built magnetic resonance imaging (MRI) system with a complete digital spectrometer has been designed for investigation of plants and animals. With the application of the latest digital integrated circuit technology, the digital spectrometer is greatly simplified without the loss of flexibility and performance. A powerful pulse sequence compiler with a graphical editor can allow the user to edit the pulse sequence more easily and more conveniently than ever before. Moreover, a permanent magnet capable of producing a 180 mm diam spherical homogeneous region is employed in our MRI system to ensure a comparatively large image size. Compared with previous work, our MRI system has the features of flexibility, relatively large imaging size, and low cost. Experimental results obtained with the proposed system are presented in this article. © 2005 American Institute of Physics. [DOI: 10.1063/1.2069707]

I. INTRODUCTION

The magnetic resonance imaging (MRI) technique has become a powerful tool for *in vivo* study of animals and plants. Although high field MRI systems have many advantages, such as high image quality and short imaging time, their application to screening large numbers of animals or plants can be prohibitively expensive. Recently, a number of groups have worked to develop low-cost MRI systems with imaging regions of small volumes by using compact low-field MR magnets.^{1–3} However, widespread use of low field MRI systems will still require lowering the cost and complexity typically associated with clinical MRI. The primary limitations of the previous work^{1,2} are small imaging size and weak pulse sequence compiling ability. Obviously, investigation of comparatively larger objects (rat and fruit) requires MRI system with larger imaging size. Moreover, a flexible spectrometer and a powerful pulse sequence compiler are indispensable to implement many imaging experiments especially in fast speed imaging. One study³ describes a complete digital MRI system based on digital signal processor (DSP). A digital rf transmitter and a receiver have been discussed in the article, but the flexibility of the spectrometer has not been mentioned. A powerful spectrometer is essential for the application of the MRI system to investigate various samples with various experimental techniques (pulse sequences). Other aspects of this previous study require further improvement. First, the magnetic field (0.1 T) is too low to produce images with high quality because the signal-to-noise ratio of MRI images depends strongly on the strength of the magnetic field. Secondly, the imaging size is relatively small (in volume of about 100 cm³), which restricts the application of the MRI system.

In this work, we have developed a MRI system which can be used to image relatively large objects. Its key features are flexibility, larger imaging size, and low cost. Nowadays almost all electronics modules for MRI systems may be commercially available, but it is still necessary to build in-house modules to lower the entire cost of MRI systems. Our in-house modules mainly include digital spectrometer, rf amplifier, and rf coils. Compared with commercial available spectrometers such as the RI spectrometer (Resonance Instruments Ltd., United Kingdom) and Tecmag spectrometer (Tecmag, Inc., USA), our spectrometer is simplified without any loss of flexibility and performance owing to the application of the latest digital integrated circuit (IC) technology. A powerful pulse sequence compiler with a graphical editor can allow the user to edit most of the clinic pulse sequences in the visualization of sequences. Furthermore, a permanent magnet capable of producing a 180 mm diam spherical homogeneous region is employed in our MRI system to ensure a comparatively large image size, which correspondingly needs a rf amplifier with larger transmitting power. With the implementation of the digital spectrometer and powerful graphical pulse sequence compiler, our system is designed as a flexible research machine which, although being low cost, would perform as well as the majority of commercial systems. In the following, we will describe the entire MRI system in detail. Finally, experimental demonstrations will be presented.

II. SYSTEM ARCHITECTURE

The block diagram of the present MRI system is shown in Fig. 1.

The magnet system consists of a permanent magnet, rf coils, and three sets of gradient coils. For geometrical reasons or for configuration reasons, a transmission/reception (T/R) switch and a low-noise preamplifier are also included in the magnet system. The spectrometer includes a personal

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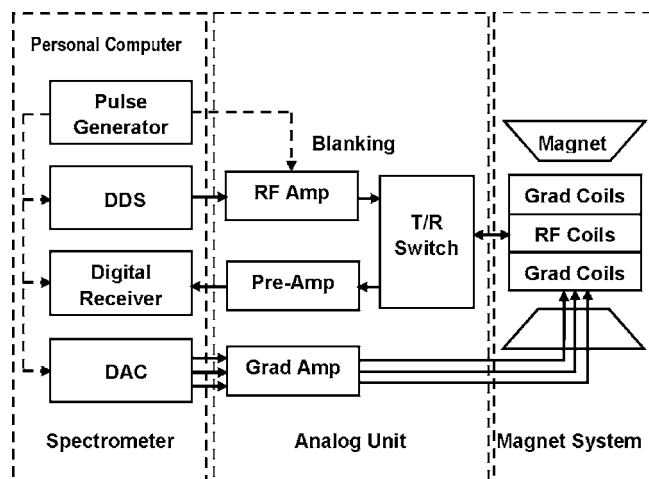


FIG. 1. Block diagram of the MRI system, - - - control lines, —signal lines. DDS—direct digital synthesis, DAC—digital-to-analog converter, rf amp—rf amplifier, pre-amp—pre-amplifier, grad amp—gradient amplifier, T/R switch—transmitter/receiver switch.

computer (PC), a pulse generator board, a rf pulse waveform generator board (rf source), a digital receiver board and a gradient waveform generator board. All four printed circuit boards can be inserted into the peripheral component interconnect (PCI) slots in the PC. The PC not only controls the spectrometer but also completes data processing and visualization. The analog unit is equipped with a rf amplifier and a gradient amplifier (three independent channels).

After pulse sequence binary information is downloaded to the onboard static random access memory (SRAM) in the digital spectrometer, the experiment will run under the control of the pulse generator. The transmitting chain⁴ starts with the rf pulse waveform generator, followed by the rf amplifier. The rf pulse waveform is first fed into the rf amplifier and then enters the rf coils through the T/R switch to excite the sample. The receive chain is initiated when the signal from the sample is received from the rf coil. The signal will be amplified by a preamplifier next to the magnet and subsequently digitized in the digital receiver to complete the digital quadrature detection. The gradient chain consists of a gradient waveform generator, a gradient amplifier, and three sets of gradient coils.

III. MAGNET SYSTEM

Although excellent electromagnet designs⁵ for imaging samples requiring a 120-mm field of view (FOV) have been available, a comparatively simple permanent magnet structure is utilized in our system due to low cost and easy maintenance. An H-shape design, displayed in Fig. 2, has been chosen, with a target field of 0.3 T and a 180 mm diam spherical homogeneous region. The gap is 220 mm and the pole diameter is 500 mm. The magnet poles were constructed using NdFeB material. The magnet is manufactured by Ningbo Jansen mechanism Corporation (Ningbo, China). The field homogeneity in a volume of about 2100 cm³ (160 mm diam sphere) measured using a nuclear magnetic resonance (NMR) gaussmeter is better than ± 20 ppm after fer-

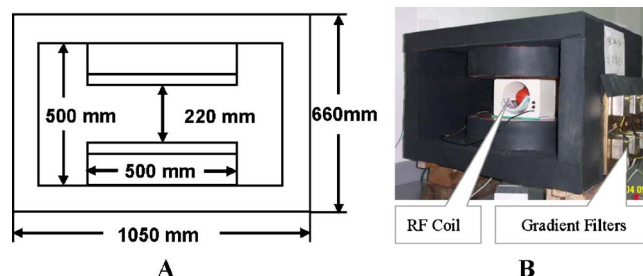


FIG. 2. (a) Schematic diagram (front view) and (b) photograph of the H-shaped permanent magnet. The magnet is about 2 tons in weight. The sample to be imaged is surrounded by a rf coil and placed in the homogeneous field region. Gradient filters are mounted on one side of the magnet frame.

roshimming, and can be further improved by applying active shimming, which is done simply by adding dc offsets to each gradient coil.

Due to the geometry of the permanent magnet, a Golay design^{6,7} for the imaging gradient coil system was employed. Because of a relatively large FOV (about 160 mm), the gradient coils were optimized by Maxwell 2D software (Ansoft, USA). The wiring patterns of the gradient coils are shown in Fig. 3 and the precise wiring positions are given in Table I. The field linearity of the optimized x , y , and z coils within the target FOV is less than (or equal to) $\pm 5.39\%$, $\pm 5.35\%$ and $\pm 2.41\%$, respectively.

The gradient coils were implemented using copper wire with diameter of about 2.5 mm. The sensitivity of each coil is somewhat different and is around 0.05 G/cm/A. They are fixed on the surface of the magnet poles. A rf shield (made from a printed circuit board) is mounted next to the gradient coils to prevent rf coupling between gradient coils and the rf coil. In addition, the preamplifier (Miteq, Inc., Hauppauge, NY, model AU-1519-7480, USA) and T/R switch were placed on the rear cover of the magnet with gradient filters placed on the opposite side of the magnet frame [see Fig. 2(b)].

A single rf coil having a multiloop geometry was employed for transmit-receive purposes. Several rf coils different in size were implemented in our laboratory to fit imaged objects. The largest rf coil, as shown in Fig. 2(b), has an inner diameter of 190 mm and length of 210 mm. Another coil displayed in Fig. 4 was constructed using four turns of copper film (0.25 mm in thickness and 16 mm in width). The rf coil is tuned in parallel with a variable capacitor and matched with another variable capacitor (see Fig. 4).

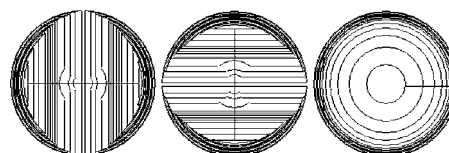


FIG. 3. Wiring patterns of three gradient coils, from left to right is x , y , z coil, respectively. The gradient coils are implemented using 13 turns of copper wire with a diameter of about 2.5 mm. Exact wiring positions are given in Table I.

TABLE I. Wiring positions of three gradient coils.

Number	Coordinate of lines (mm) X coil	Radius of arcs (mm) X coil	Coordinate of lines (mm) Y coil	Radius of arcs (mm) Y coil	Radius of circles (mm) Z coil
1	6	240	6	240	240
2	26	236	26	236	236
3	42	233	44	233	233
4	61	230	63	230	230
5	85	226	84	226	226
6	92	222	92	222	222
7	100	219	100	219	218
8	109	215	109	215	212
9	118	212	118	212	202
10	126	208	126	208	185
11	145	205	148	205	160
12	164	201	170	201	122
13	183	198	180	198	66
14		80		78	
15		50		49	
16		32		32	

IV. SPECTROMETER

The spectrometer consists of a pulse generator board, a rf waveform generator board, a digital receiver board, and a gradient waveform generator board. A standard Pentium-IV (Intel, Inc., USA), 2400 MHz PC with 512 MB of RAM was used to control the system. The spectrometer is constructed by using a number of IC chips. All four printed circuit boards based on the PCI bus are equipped with PLX9052 (Ref. 8) (PLX, Inc., USA) chips which serve as a bridge between PCI bus and the onboard local bus. A field programmable gate arrays (FPGA) chip [XC2S200 (Ref. 9), Xilinx, Inc., USA] is utilized for logic control in the above four boards. The application of FPGA chip simplifies the design of the spectrometer. In addition, one or two 256K $\times$ 16 bit SRAM chips [CY7C1041B (Ref. 10) CYPRESS, Inc., USA] are attached to each board for storing data, such as the event table, rf waveform, gradient waveform, and the sampled MRI data.

The precise timing of a pulse sequence is controlled by the pulse generator. The principle of our pulse generator is similar to that of the pulse generator mentioned in a previous article,¹¹ but we simplify the design by using a FPGA chip for logic control instead of many discrete digital devices.

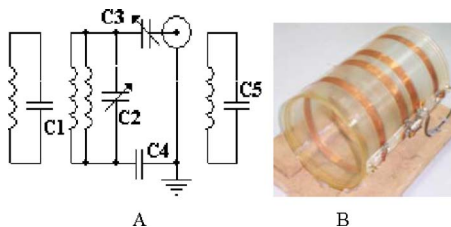


FIG. 4. (a) The circuit diagram and (b) photograph of the rf coil. In the circuit diagram, $C1=C5=420$ pf, $C2=435\text{--}640$ pf, $C3=55\text{--}101$ pf, $C4=200$ pf. The $C2$ capacitor is used to tune and the $C3$ capacitor is for matching. The rf coil with 220 mm length (Plexiglas skeleton) and 150 mm inner diameter is constructed using 4 turns of copper film (0.25 mm in thickness and 16 mm in width). The gaps between adjacent copper films are 40 mm, 14 mm, 40 mm from left to right.

Our pulse generator has 16 independent channels and each channel is used to control an event, such as rf updating (amplitude, frequency, and phase), acquisition triggering, and rf gating, etc. The state of each channel is identified by one binary bit (0, low level; 1, high level). The states of the 16 channels are defined as “events” within a “delay” period. The timing resolution of 20 ns is realized by using a 50 MHz clock. The minimal pulse width is 100 ns. Before running a pulse sequence, an event table (array) with the “events-delay” structure generated by the pulse compiler is downloaded to the SRAM through the PCI bus. The control logic in FPGA will read each element in the event table in order and output the corresponding pulses to the current element.

A direct digital synthesis (DDS) chip [AD9854 (Ref. 12) Analog Device, Inc., USA] is utilized on the rf pulse waveform generator. The AD9854 digital synthesizer is a highly integrated DDS device. The maximum output frequency is 135 MHz when referenced to the 300 MHz system clock. It has the ability to rapidly manipulate the instantaneous amplitude, frequency and phase. The rf pulse waveform generator is made with some modifications and improvements of the old DDS design.¹³ The main modification is that two SRAM chips are added to the printed circuit board for storing the predefined rf pulse waveforms. In addition, an industry standard architecture (ISA) interface is replaced with PCI interface. Each point of the rf pulse waveform is determined by parameters of the amplitude/frequency/phase. Once FPGA receives a rising edge signal from the pulse generator, the control logic in FPGA will transfer current rf data (amplitude-frequency-phase) from SRAM to DDS register and update the DDS output. The minimum switching time between adjacent rf waveform data are 1 μ s.

The digital receiver is based upon the software-radio technology [Analog Devices’ AD9874 (Ref. 14) chip]. The block diagram is shown in Fig. 5.

The AD9874 is a general-purpose IF subsystem that digitizes a low level 5–300 MHz IF input with an output

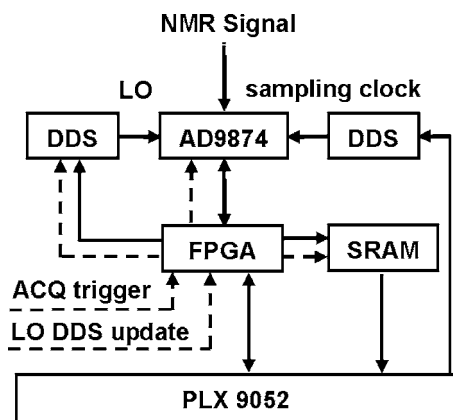


FIG. 5. Block diagram of the digital receiver card, - - - control lines, —data/signal lines. Two DDS are employed, respectively, for local oscillator (LO) and sampling clock. The rising edge signal from the ACQ trigger line will force FPGA to control AD9874 sampling. And the rising edge signal from the LO DDS update line will update the LO DDS register with the current LO's amplitude, frequency, and phase value stored in FPGA. All the acquired data are buffered in SRAM (1 Mbyte). These data will then be transferred to the PC through PCI bus after completing acquisition.

signal bandwidth ranging from 6.8 to 270 kHz. The signal chain of the AD9874 consists of a low noise amplifier, a mixer, a bandpass sigma-delta analog-to-digital converter (ADC), and a decimation filter with programmable decimation factor. The utilization of the AD9874 chip greatly simplifies the digital receiver.

Two DDS chips are employed, respectively, for the local oscillator (LO) and sampling clock of AD9874 (see Fig. 5). The acquisition trigger line and the LO DDS update line are directly connected to the FPGA (also see Fig. 5). Once the FPGA receives a trigger signal, the corresponding state machine constructed in the FPGA will carry out the specified action. All the acquired data are buffered in SRAM (1 Mbyte). After receiving the interrupt signal emitted by the digital receiver, the PC will immediately read acquisition data from the SRAM on board.

Only 16 discrete values of the spectral width are provided by the AD9874 chip through change of the decimation factor. This is of course inadequate for most of the MRI experiments. Since the spectral width not only depends on the decimation, but also on the frequency of the sampling clock, we may further change the spectral width by setting the sampling clock. Therefore, a DDS chip is utilized to serve as the sampling clock of the AD9874 chip for convenient variation of the sampling clock. As a result, the spectral width can be coarsely set by the decimation, and determined finely by the ADC clock. In this way the spectral width may be adjusted with very high precision.

The gradient waveform generator utilizes one 16-bit digital to analog converter (DAC) chip (AD5542, Analog Device, Inc.) for each gradient channel. The predefined gradient waveform data are stored in the onboard SRAM. When FPGA receives a rising edge signal from the pulse generator, the RAM RD/WR logic control block implemented by the FPGA will transfer the current waveform data from SRAM to DSP block constructed in FPGA for the real-time pre-emphasis algorithm. Then the pre-emphasized waveform data are sent to DAC. Pre-emphasis is based on a finite im-

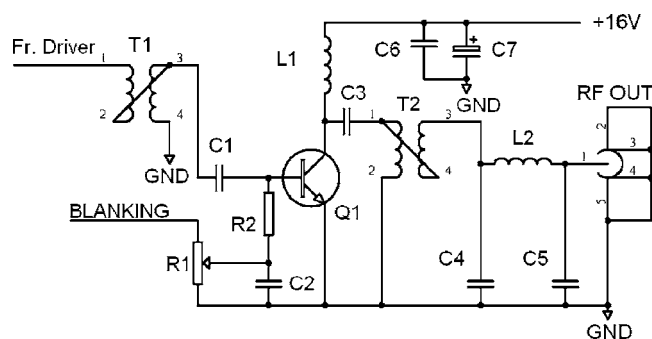


FIG. 6. Schematic of the final rf power amplifier. In the circuit diagram, $T1=T2=10\text{ T}$, $R1=1\text{ K}$, $R2=51$, $L1=2.7\text{ uH}$, $L2=610\text{ uH}$, $C1=C2=C3=10\text{ nf}$, $C4=C5=240\text{ pf}$, $C6=0.1\text{ uf}$, $C7=2200\text{ uf/25V}$, $Q1=C2630$. The amplifier works in class A (class C) when the blanking signal goes high (low).

pulse response (FIR) model which involves the exponent parameters of amplitude and time constants.^{15,16} Parameters may be adjusted according to the gradient field. The differential output is adjusted to $\pm 5\text{ V}$.

The spectrometer (one PC and four PCI boards) discussed above has some important features. First, the spectrometer structure is greatly simplified while keeping its flexibility. Second, it is a complete digital spectrometer, including frequency synthesis, receiver detection and gradient waveform pre-emphasis that ensures high overall performance. Third, our spectrometer operates in a broad frequency range (5–135 MHz for transmission/reception). Finally, both transmission and reception channels can be fully controlled by the pulse sequencer. That is, the amplitude, the phase, and the frequency of the rf signals can be switched very fast ($1\text{ }\mu\text{s}$) by the pulse generator. As will be seen later (discussed in Sec. VII), this feature is of great importance for the simplifying the implementation of the complex pulse sequences.

V. ANALOG UNITS

The rf pulses from the transmission channel of the rf pulse waveform generator are then fed into the rf transmitter, which consists of a rf driver and a rf power amplifier. The rf signal is first amplified up to about 2 W in the driver. The output of the driver is then used to drive the final power amplifier. Figure 6 displays a schematic diagram of the rf power amplifier (final stage) employed in this MRI system.

The rf amplifier is of standard design and operates in class A when the blanking signal is high. A power transistor C2630 (Mitsubishi Semiconductors, Japan) is utilized. The output impedance of the transmitter is adjusted to $50\text{ }\Omega$. The final maximum output power is approximately 16 W, giving the shortest duration of 0.8 ms for a three-lobe SINC soft pulse (90°) when the largest rf coil is employed. The rf pulses from the transmitter are fed into a T/R switch.¹⁷ The main functions of the T/R switch are protection of the pre-amplifier from damaged high rf voltages during rf pulsing as well as isolation of noise of the rf power amplifier from the rf coil during signal sampling.

Many types of commercially available audio amplifiers can be utilized for use as the gradient amplifiers with slight

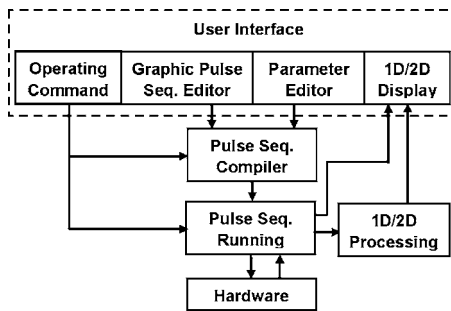


FIG. 7. Structure chart of the software. Here Seq is an abbreviation for Sequence.

modifications. In our MRI system, we use two commercially available stereo-audio amplifiers from Yamaha Corporation (P3500S, Hamamatsu, Japan), which are able to drive 20 A or more with a $2\ \Omega$ resistive load. At final output, a $2\ \Omega/25\ \text{W}$ resistor is used in series with the gradient coil to avoid oscillation of the amplifier. In order to extend the performance of the audio amplifier down to dc, it is necessary to make simple modifications, including the change of the capacitors C451 and C455 (dc isolation capacitors that are easily found on the printed circuit board) to 1000 and 4700 Ω resistors, respectively. It also may be necessary to increase the threshold of protection in the amplifiers if one wants to operate the gradients with high duty cycle.

VI. SOFTWARE

The software WinMRI was developed under Windows XP using Delphi 7.0 (Borland, USA). Figure 7 illustrates the functional block diagram of the software.

The user interface consists of four blocks: (1) operating command; (2) the graphical pulse sequence editor; (3) the parameter editor; (4) 1D/2D display.

A simple 2D gradient echo (GRE) pulse sequence is taken as an example for describing the graphical pulse sequence editor. The GRE sequence is given in Fig. 8 while the graphical sequence editor interface with the GRE sequence is shown in Fig. 9.

As can be seen in Fig. 9, the GRE sequence edited in the graphical sequence editor is similar to that in Fig. 8. The user can draw the pulse sequence in the graphical editor instead of the programming pulse sequence in text sequence editor.

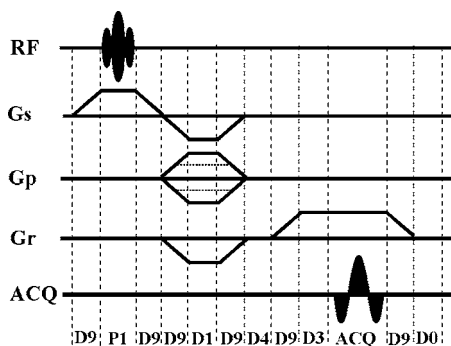


FIG. 8. A simple gradient echo sequence. This sequence is divided into 12 events which are individually assigned by delay time indicated in the figure.

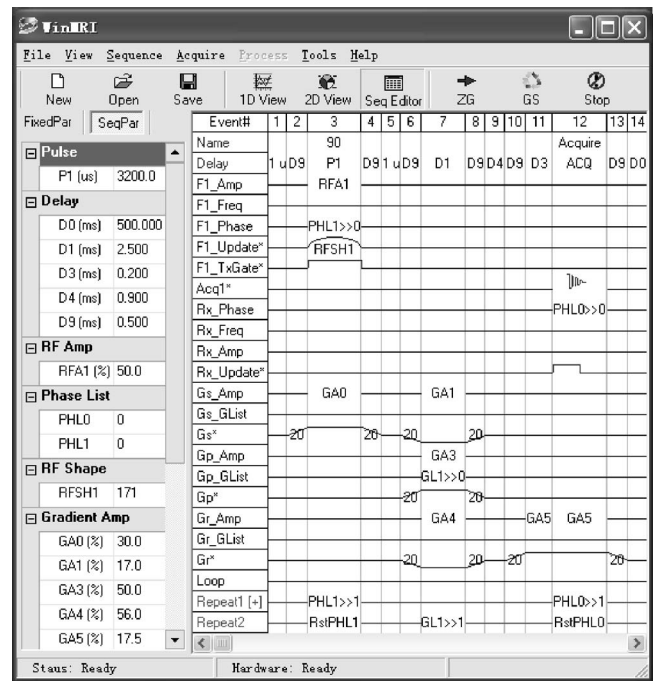


FIG. 9. Graphical sequence editor with gradient echo (GRE) pulse sequence. Left: parameter editor; Right: graphical sequence editor. The GRE sequence edited in graphical editor is similar with that in Fig. 8. The various icons represents a different child object. The properties are indicated by parameter names. For example, event 3# will simultaneously trigger a soft pulse indicated by RFSH1 and a slice-selective gradient whose amplitude is indicated by GA0. The amplitude and phase of the soft pulse is indicated by RFA1 and PHL1, respectively. Two repeats are utilized in the GRE sequence. Repeat1 is for signal average with phase cycling and repeat2 is for gradient phase-encoding. The pointer of GL1 list is moved down one position in repeat2. The parameter editor only shows the parameters appearing in the graphical editor. The user can easily set the parameter according to the pulse sequence in the graphical editor.

A MRI experiment evolves many basic sequence objects, such as rf (transmission, assigned to F1 channel further), receiver (acquisition, Rx), and gradients (slice gradient Gs, phase encoding gradient Gp and read gradient Gr), etc. They are assigned to different properties according to their corresponding hardware structures. For instance, the rf has the properties of amplitude, frequency, and phase.

The user can develop almost all types of MRI pulse sequences (allowing for the visualization of sequences) using the graphical sequence editor. Pulse sequences are created by adding “events.” The events are the numbered columns that appear in the sequence editor. The rows in the sequence editor represent the sequence objects mentioned above. All the objects that are active during an event occur simultaneously during that event and for the duration that is indicated in the “delay” line. For convenience, some sequence objects are further divided into several child objects. For example, hard and soft rf pulses are derived from the rf object. The child objects are represented by icons, whereas the properties are indicated by parameter names and parameter identifiers (ID). The parameter name is displayed in the interface of the pulse sequence editor and the parameter editor. The parameter ID is just used for speeding up the parameter indexing procedure. Each parameter name has a unique entry which is allowed to modify the parameter value represented by the cor-

responding parameter name in the parameter editor. The parameter is divided into three types: variable, list, and shape. The variable (scalar) specifies to single value parameters, such as delay time parameters (identified by D0–D31), pulse time parameters (P0–P31), rf amplitude parameters (RFA0–RFA15), gradient amplitude parameters (GA0–GA31), etc. The list (vector) and the shape basically contain an array of values. The list parameters include the rf amplitude list parameters (RFAL0–RFAL15), rf frequency list parameters (FQL0–FQL15), rf phase list parameters (PHL0–PHL15), gradient phase encoding list parameters (GLO–GL15), and variable delay list parameters (VDL0–VDL15), while the shape parameters consist of rf shape parameters (RFSH0–RFSH15) and gradient shape parameters (GSH0–GSH15). In order to edit a complex pulse sequence, the list also supports the arbitrary skip of the list pointer. For example, GL1 $\gg$ 1 means first using the current value of the GL1 list and then moving down one position from the current position of the GL1 list. The rf pulse waveform and the gradient waveform are ASCII files. For the sake of convenience, many conventional rf pulse waveforms or gradient waveforms have been generated and stored under the shape directory of the WinMRI software. Some frequently used lists are saved as ASCII files under the list directory. In addition, some standard pulse sequences are also stored under the sequence directory.

Operating commands mainly include fundamental operations for data acquisition and processing. In addition, the 1D/2D display window is also the essential part which can allow the user to view a 1D or 2D plot of the MRI image. Particularly, the 1D view displays the FID signal in real-time mode during acquisition.

The pulse sequence edited by the user is directed to the pulse sequence compiler module made up of syntax check and compilation submodules. Syntax check mainly verifies the pulse sequence and validates the associated parameters. Compilation is divided into three steps. First, it translates the graphical sequence element of each row into internal information tables. Four information tables, including event information table, rf information table, gradient information table, and receiver information table, are defined in the program. Second, the target code specific to the relevant spectrometer hardware for the first repetition is generated according to the four information tables and parameter values in the parameter editor. Third, the target code generated in the second step is partially modified for each repetition due to processing the different skip of list pointer in each scan repeat. The first two steps are done before executing the pulse sequence in order to accelerate the third compilation. This initial compilation spends about 50 ms. The third step in the compilation process (frequently less than 1 ms) is repeated (according to the list skip patterns in the repetition) for each repetition of the scan. To achieve the shortest possible repetition (TR) the third step compilation for the next repetition begins as soon as the current acquisition of the pulse sequence is started. Furthermore, only the modified binary data is downloaded to the hardware except for the first repetition.

The main functions of the pulse sequence running mod-

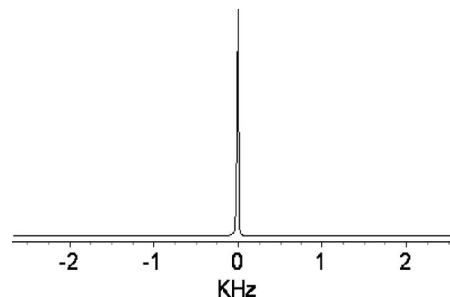


FIG. 10. Single-hard-pulse ^1H spectrum of 1% CuSO_4 water obtained with the following parameters: 90° rf pulse width $P1=180\ \mu\text{s}$, dead time $D3=100\ \mu\text{s}$, spectral width $\text{SW}=5.0\ \text{kHz}$, number of sampling $\text{TD}=1\ \text{K}$, number of excitations $\text{NEX}=1$.

ule are data updating for the hardware and real-time display of the acquired data in the current excitation.

After scanning the pulse sequence, the entire acquisition data can be viewed in 1D or 2D style. Moreover, the software provides a COM/DCOM programming interface that allows the client program to control or control the WinMRI software remotely and conveniently.

The 1D process module includes some routine processes, such as zero filling, apodization, 1D fast Fourier transform (FFT), phase correction, baseline correction and integration. The 2D process module provides zero filling, apodization, 2D/3D FFT, and window width/level adjustments. It will take about 6 s for performing a 3D FFT with data size $256 \times 256 \times 16$ and displaying the data on screen. The MRI raw data can be exported for further processing if necessary.

VII. EXPERIMENTAL RESULTS

To demonstrate the performance of the proposed MRI system, a number of conventional MRI experiments were carried out in the following. The experimental parameters are as follows: ^1H resonant frequency $\text{SF}=12.73\ \text{MHz}$ and offset $\text{O1}=10100\ \text{Hz}$. The maximum current of three gradient coils is adjusted to 20 A. Gradient dc offsets ($I_{\text{offsetx}}=48.8\ \text{mA}$, $I_{\text{offsety}}=-9.0\ \text{mA}$, $I_{\text{offsetz}}=30.5\ \text{mA}$) utilized to shim the main static magnetic field are always applied to three gradient coils and the rf coil (shown in Fig. 4) is employed in all the experiments. In addition, gradient waveform pre-emphasis for three axes is applied before imaging experiment.

Figure 10 shows the typical ^1H spectrum obtained with a 100 mm diam ball filled with water (doped with 1% CuSO_4). The linewidth at half-height is about 1.5 ppm. This line broadening is due mainly to the high-order residual field inhomogeneity which cannot be compensated by the application of dc offsets.

Multislice MRI imaging is a routine clinical technique. For conventional analog spectrometers, slicing can be accomplished by using an additional B_0 coil. But the B_0 coil may also produce serious eddy currents that will pollute the images. For digital spectrometers where DDS serves as the rf source, slicing may be done by switching the rf frequency instead of changing the magnetic field strength. In this situation, the rf transmitter and the rf receiver must be synchronized throughout the experiment. The synchronization is achieved by a so-called phase rewinding technique.¹⁸ But it

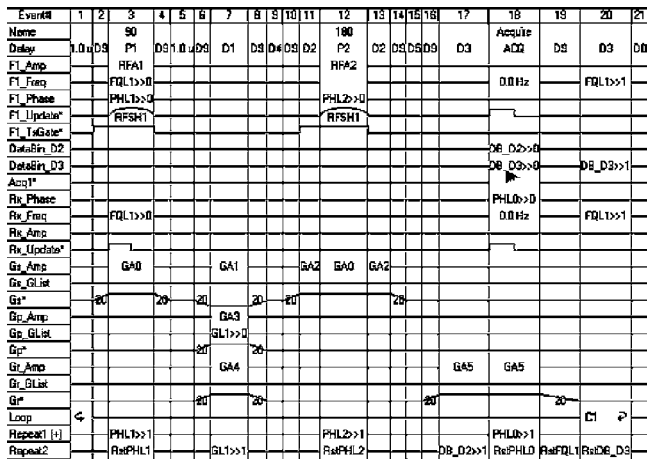


FIG. 11. Multislice spin echo pulse sequence edited in the graphical sequence editor. As can be seen in the event 3#, the frequency of the transmitter and the receiver are simultaneously switched to accommodate nine slice frequencies. In the event 18#, the frequency of the transmitter and the receiver are restored to the receiver's frequency. After each slicing, the pointer of the FQL1 list is moved down one position, which is indicated in event 20#.

requires precise timing adjustment of the pulse sequence. Because the LO of the digital receiver is provided by a DDS, we can switch the phase and the frequency of the receiver. In this case, the synchronization may be achieved simply by simultaneously switching the frequency of the transmitter and the receiver, as shown in Fig. 11. This feature is also useful in implementation of other complex pulse sequence such as fast spin echo (FSE) and echo planar imaging (EPI).

Figure 12 depicts multislice 2D images (proton density weighted images) of a watermelon (the largest diameter is 120 mm, approximately) obtained with the spin-echo pulse sequence shown in Fig. 11. A sinc-shaped rf pulse with three lobes is used in this pulse sequence.

Finally in Fig. 13 we show multislice images of a rat obtained with the 3D gradient echo (3D GRE) pulse se-

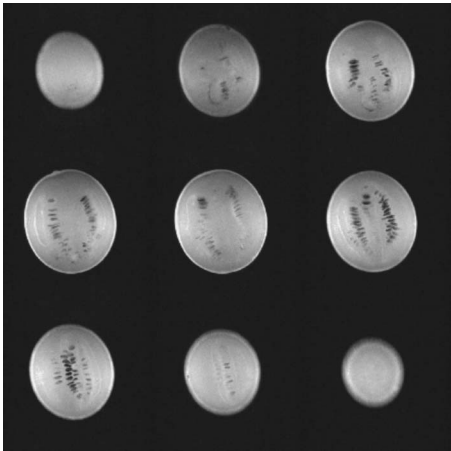


FIG. 12. Multislice spin echo images of a watermelon (the largest diameter is 120 mm, approximately). The experiment parameters are as follows: TR/TE=1000/26 ms, number of excitations NEX=2, acquisition matrix size is 256×192, field of view FOV=150×150 mm, resolution is 0.59×0.78 mm and slice thickness=8 mm. Total acquisition time was 6.4 min. (No postprocessing is performed.)

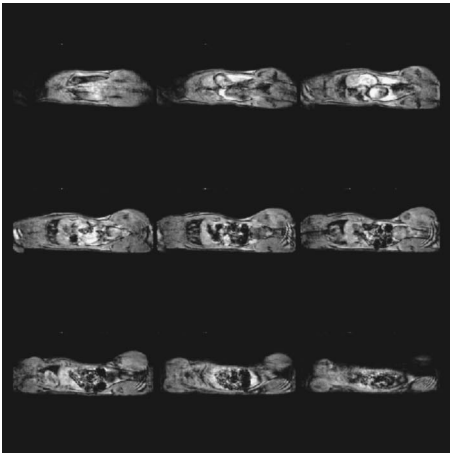


FIG. 13. Coronal MRI images of mouse obtained with conventional 3D gradient echo. The rat was 210 mm in length and about 700 g in weight. Only 9 central slices selected from the acquired data are displayed. The experimental parameters are as follows: TR/TE=161/12.8 ms, acquisition matrix size for each slice is 256×192, number of phase encoding step in the third dimension (slicing) is 16, number of excitation NEX=1, field of view FOV=150×150×75 mm, rf flip angle=30°. Total acquisition time is 8.2 min. (No postprocessing is performed.)

quence. The rat was studied under a protocol approved by our institution's research animal resources center. It was anesthetized (20–30 mg/kg amobarbital).

The rat is about 210 mm in length (not including the tail) and about 700 g in weight. Seven slices in the edges are deleted from the figure due to their lack of information.

We found experimentally that the image quality derived from the spin echo sequence is some what better than those obtained with the gradient echo method. This is due mainly to the inhomogeneity of the magnetic field presented in the MRI system in spite of active shimming.

So far we have demonstrated and verified the performance and flexibility of the MRI system proposed in this article. One of the important characteristics of the MRI system is the application of digital technology that dramatically simplifies the design of the spectrometer without scarifying the performance. In addition, a powerful pulse sequence compiler with a graphical editor can allow user to edit the pulse sequence more easily and more conveniently than ever before. We have also tried the pre-emphasis of the applied gradient waveforms, even when there is little effect on the conventional spin-echo or gradient-echo sequence. The reduced eddy current effects are due to, we guess, the smaller size of our magnet compared with that in whole-body clinical MRI systems. However, it is significant for fast imaging sequences such as RARE or EPI. This part of the work will be described later in a separate article. Compared with most of the previous work, our system can explore relatively larger objects and can generate images with much better quality. The entire system may be implemented at the cost of less than \$12 000 in our laboratory. As a result, the MRI system presented in this article may find many applications in the animal and plant investigations. No doubt it can also be applied to explore other objects provided that the samples can be fit into the homogeneous region of the magnetic field.

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